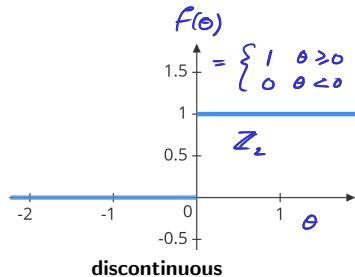
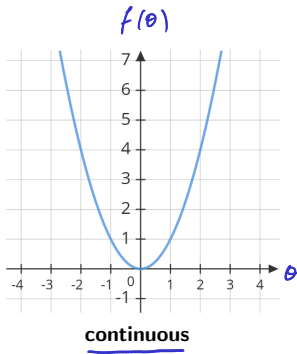
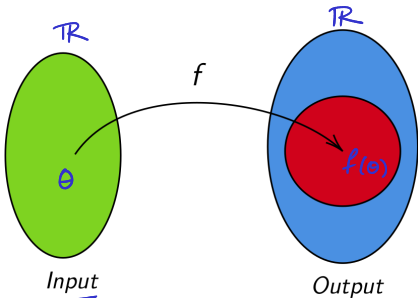


Functions on sets



$$0 = f(0-\epsilon) \neq f(0+\epsilon) = 1$$

$$\epsilon \rightarrow 0$$

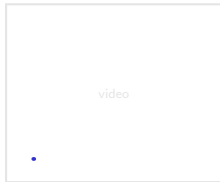
Function f from A to B

Input from A (domain of f)

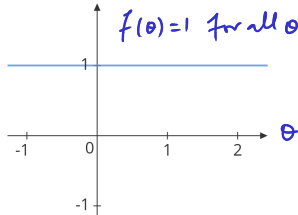
Output in B (codomain of f)

$$f: \underset{\mathbb{R}}{A} \rightarrow \underset{\mathbb{R}}{B}$$

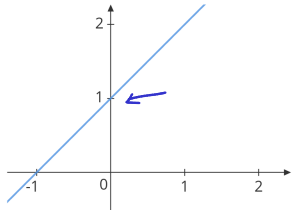
$$\{f(a)\} = \text{Range } f(A) \subseteq \underline{B}$$



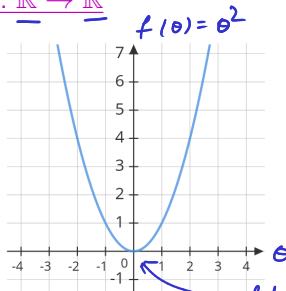
Real-valued continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$



constant

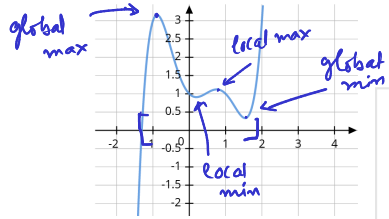


linear



quadratic

global min ==



higher-degree polynomial

$f(\theta) = \theta^5 - 2\theta^4 - \theta^3 + 3\theta^2 - \theta + 1$

